



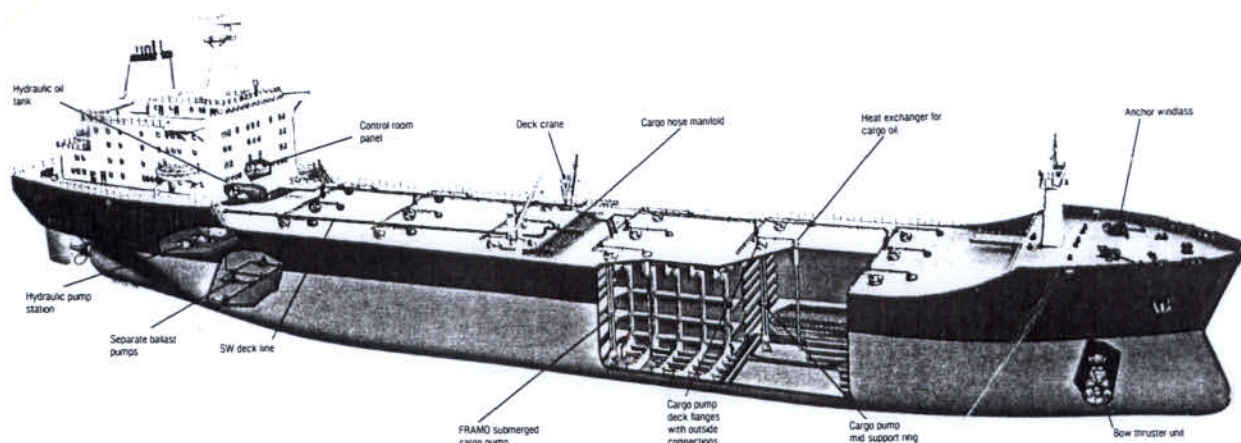
Frank Mohn Services AS

No.5 - 1991

Service Bulletin

MAKE YOU UP TO DATE WITH THE LATEST NEWS FROM FRANK MOHN SERVICES AS

Operation of FRAMO Cargo pumping system



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DEAR CUSTOMER!

FRAMO cargo pumping systems are all very easy to operate.

We have, however, seen a difference both in operational and maintenance costs from one vessel to another.

In order to minimize these costs, we would like to highlight a few important operational issues.

LOADING:

Always start the loading through the cargo pump slowly, with the cargo valve partly open. This to avoid any pressure peaks in the pump housing.

When the cargo line is completely filled, open the cargo valve to max. open position.

NB! Remember max. pressure at top cover plate is 8 bar.

DISCHARGING:

- Start each pump against closed cargo valve and leave it running at 30-40 bar for 2-3 min. before increasing the speed.
- Raise the pump pressure to above shore line pressure, before opening the cargo valve. This to avoid "back flow" (over pumping).
- After starting several cargo pumps - if the manifold pressure does not raise - it is no use to start another cargo pump.
- Regulate the hydraulic system pressure to approx. 15 bar above the highest consumer pressure (f.ex. cargo pumps).

STRIPPING:

It is essential for the stripping result that the cargo valve does not leak.

It is also important to find the correct speed and that the pumps wear rings are in good condition.

Pump speed during stripping depends on discharge head and differs from ship to ship.

Ideal stripping speed must therefore be based upon experience on board

MAIN RULE: Always strip against lowest possible back pressure.



TANK WASHING:

Reduce pump speed to match tank cleaning machines capacity.

Avoid dry running of cargo pump. Remember to reduce hydraulic system pressure to 15 bar above the highest consumers hydraulic pressure.

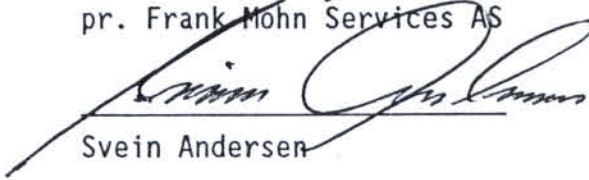
PREVENTIVE MAINTENANCE:

- Purge the pumps cofferdam before and after using the pump.

NB! Make sure that the cargo pumps cofferdam is open, -
i.e. air is coming out of the exhaust trap.

- If air is used as purging medium, drain airline for possible water condensation regularly.
- Deairate the hydraulic system on a regular basis. Avoiding air in the system will increase the components lifetime.
- Check the pressure drop over the filters.
- Take oil sampling every 3 rd month, including particle counting ISO 4406.
- Check the cargo pumps condition against the performance curve once a year.
- Never start dismantling of any cargo pump or other component in the system before you know if - what - and where you have a problem.

Yours faithfully
pr. Frank Mohn Services AS



Svein Andersen